



Cluster-based genetic segmentation of time series with DWT

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ABSTRACT

A time series is composed of lots of data points, each of which represents a value at a certain time. Many phenomena can be represented by time series, such as electrocardiograms in medical science, gene expressions in biology and video data in multimedia. Time series have thus been an important and interesting research field due to their frequent appearance in different applications. This paper proposes a time series segmentation approach by combining the clustering technique, the discrete wavelet transformation and the genetic algorithm to automatically find segments and patterns from a time series. The genetic algorithm is used to find the segmentation points for deriving appropriate patterns. In fitness evaluation, the proposed approach first divides the segments in a chromosome into k groups according to their slopes by using clustering techniques. The Euclidean distance is then used to calculate the distance of each subsequence and evaluate a chromosome. The discrete wavelet transformation is also used to adjust the length of the subsequences for calculating the similarity since their length may be different. The evaluation results are utilized to choose appropriate chromosomes for mating. The offspring then undergo recursive evolution until a good result has been obtained. Experimental results show that the proposed approach can get good results in finding appropriate segmentation patterns in time series.

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1. Introduction

Time series has always been an important and interesting research field due to its frequent appearance in different applications. Many research topics, including anomaly detection (Keogh, 2005; Keogh et al., 2005; Liu et al., 2008), similarity measurement (Han et al., 1999; Lian and Chen, 2008), dimension reduction (Huang and Yu, 1999; Lendasse et al., 2001) and segmentation (Cohen and Adams, 2002; Guralnik and Srivastava, 1999; Himberg et al., 2001; Keogh et al., 2001; Liu et al., 2008; Oliver et al., 1998; Wang and Wang, 2004), are greatly related to it. Some domains such as bioinformatics (Ben-Dor et al., 1999; Aach and Church, 2001; Choong et al., 2009; Erdal et al., 2004), medical treatment (Valente and Chavarrias, 2000; Guýguin et al., 2008) and finance (Chang and Fan, 2008; Chung et al., 2004) especially emphasize it for making good prediction and decision.

Time-series segmentation is an important processing procedure for analyzing a time series. Many approaches have thus been proposed in previous studies. Guralnik and Srivastava (1999) proposed an event detection algorithm to detect events from time series.

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Himberg et al. (2001) proposed a context recognition algorithm to recognize significant activities of mobile devices. Bellman proposed a dynamic programming approach for searching for proper segmentations with similar mean values and different variance values (Bellman, 1961). Wang and Wang (2004) proposed a structure-adaptive k piecewise segmentation approach to represent a time series by several piecewise lines. It first found the local minimum and local maximum points of a time series. Regression lines were then derived from these points and used to represent the segmentations. Some other approaches like neural networks (Fancoua and Principe, 1996; Feng et al., 2005) and mathematical models (Penny and Roberts, 1999) were also used for time-series segmentation.

The segmentation problem is usually converted into an optimization problem. Recently, the genetic algorithm has been proven to be very suitable for solving optimization problems. Chung et al. (2002, 2004) and Fu et al. (2001) thus proposed several GA-based approaches to segment time series. They allowed users to specify a set of desired patterns and search for them during the segmentation processing steps. They used the real-number schema to encode each chromosome, which represented a possible segmentation result. They also defined perceptually important points (PIP) and used them to adjust the lengths of the segments to more match the predefined patterns. The lengths of the segments could thus become more flexible. The approach, however, needed domain knowledge for predefined patterns.

In this paper, we propose a time-series segmentation approach by combining the clustering technique, the discrete wavelet transformation and the genetic algorithm to automatically find the proper segments and appropriate patterns from a time series. The genetic algorithm is used to find the segmentation points for deriving appropriate patterns under some criteria. The proposed approach first divides the segments (subsequences) in a chromosome into k groups according to their slopes by using the k -means clustering technique (McQueen, 1967). The Euclidean distance is then used to calculate the distance of each subsequence and evaluate a chromosome. Because two segments may have different lengths, the discrete wavelet transformation is used to make the lengths of two segments equal such that the distance can be calculated. The evaluation results are then utilized as measures for choosing appropriate chromosomes for mating in the next generation. The offspring thus undergo recursive evolution until a good segmentation result has been obtained. Experiments on a real dataset are also made to demonstrate the effectiveness of the proposed approach. The proposed approach thus has the following advantages.

1. Not only segmentation results can be obtained but also appropriate patterns can be automatically derived from the time series via the combination of the genetic algorithms and the clustering techniques. The property is very useful for further analysis such as common pattern analysis from financial data.
2. Derived patterns can easily be scaled up via the discrete wavelet transformation. The property makes derived patterns more flexible and more diverse. It may help experts easily find their desired interesting results for further analysis.

The remaining parts of this paper are organized as follows. The adjustment process for the time-series segmentation is explained in details in Section 2. The proposed algorithm is described in Section 3. Experiments to demonstrate the effectiveness of the proposed algorithm are described in Section 4. Conclusions and future works are given in Section 5.

2. Details of the proposed approach

2.1. Chromosome representation

Two basic encoding methods have commonly been used in genetic algorithms. One is bit-string schema and the other is real-number schema (Michalewicz, 1994). In this paper, the real-number schema is adopted to encode chromosomes. A time series with n data points can be represented as $T = \{d_1, d_2, \dots, d_k, \dots, d_n\}$, where d_k denotes the k th data point in the time series T . A segmentation result can be represented by several segment points. A chromosome in the proposed approach is thus composed of the segment points for certain segmentation. The chromosome representation for a segmentation result is shown in Fig. 1.

In Fig. 1, each individual is represented as a real-number string with the alphabet of $\{c_1, c_2, \dots, c_h, \dots, c_p\}$, where c_h means that the

cut point lies between the two data points d_{ch} and d_{ch+1} . Besides, the following constraint must be satisfied: $c_1 > c_2 > \dots > c_p$. Thus, the first segment S_1 is from d_1 to d_{c_1} , the second segment S_2 is from d_{c_1+1} to d_{c_2} , and so on. The last segment S_{p+1} is from d_{c_p+1} to d_n . Below, a simple example is given to illustrate the chromosome representation.

Example 1. Assume a time series with twenty data points is given as shown in Table 1. Assume there is a chromosome C_1 encoded as $\{4, 8, 12, 16\}$. It thus represents the segmentation result shown in Fig. 2 according to the encoding scheme.

The chromosome C_1 has four cut points, which are 4, 8, 12 and 16. It thus has five segments, namely S_1, S_2, S_3, S_4 and S_5 , as shown in Fig. 2. The first segment S_1 contains four data points ($=4-0$), which are 8, 12, 12, 2. The second segment S_2 contains four data points ($=8-4$), which are 2, 4, 6, 8. The other segments can be similarly derived. Each chromosome thus represents a possible segmentation result. This representation can allow genetic operators to search for appropriate solutions.

2.2. Initial population

A genetic algorithm requires a population of feasible solutions to be initialized and updated during the evolution process. As mentioned above, each individual within a population is a possible segmentation result for a given time series. An initial set of chromosomes is thus generated with some constraints to form feasible segments. In this paper, an initial population is generated by Chung's approach (Chung et al., 2004). n cut points are first generated such that each segment has equal length. Variance values are then added to the cut points to move them left or right to form initial chromosomes.

2.3. Clustering segments into groups

The proposed approach hopes not only to segment time series, but also to derive appropriate patterns from given time series simultaneously. The clustering technique is used to achieve this purpose. Some good works on clustering time series have been proposed (Chen, 2005; Keogh et al., 2003). The k -means clustering approach is used here to divide segments in a chromosome into k groups. Thus, appropriate attributes for clustering segments need to be first found.

Assume there are two segments Q and S , each of which consists of n data points. Let $Q = \{q_1, q_2, \dots, q_n\}$ and $S = \{s_1, s_2, \dots, s_n\}$, respectively. For evaluating the similarity of two segments, the Euclidean

Table 1

A time series with twenty data points.

A time series
8, 12, 12, 2, 2, 4, 6, 8, 8, 6, 4, 2, 2, 4, 4, 2, 2, 4, 4, 2

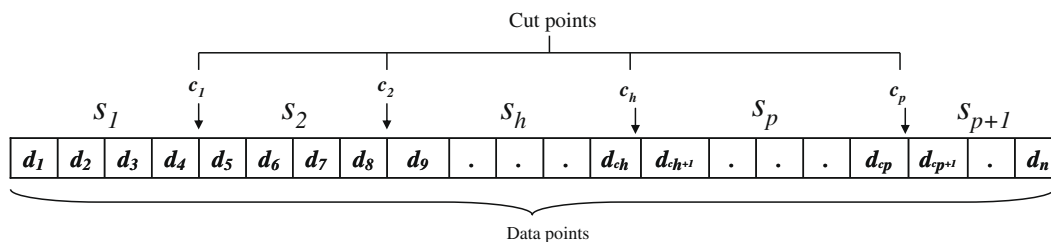


Fig. 1. A chromosome representation.

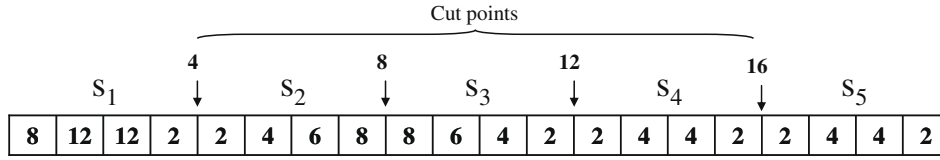


Fig. 2. The segmentation result according to the cut points in the chromosome C_1 .

distance ($ED_{DPoints}$) of data points in segments (also represented as $Dist(Q, S)$ in the later section) is the criterion most commonly used. It is defined as follows:

$$ED_{DPoints}(Q, S) = \frac{\sqrt{\sum_{k=1}^n (q_k - s_k)^2}}{n}, \quad (1)$$

where Q and S are two segments in a chromosome, n is the length of the segments, q_k and s_k are the k th data points in Q and S , respectively. In our approach, the length of two segments in a chromosome may be different, such that using data points as attributes for clustering segments are not appropriate. In this paper, we thus use slopes of segments as attributes to cluster segments. A parameter $numSlopes$, which represents the number of slopes in each segment, is first set by users. $(numSlopes + 1)$ data points are then picked from each segment to form the $numSlopes$ slopes. The first and the last points are the two in the ends. The other $(numSlopes - 1)$ data points are systematically selected such that every two adjacent data points have equal length. A slope value is then derived from two adjacent data points. Let the slopes extracted from a segment S be denoted as $Slp_1^S, Slp_2^S, \dots, Slp_{numSlopes}^S$. The r th slope value (Slp_r^S) of S is calculated by the r th and the $(r + 1)$ th selected points in S . Formally, let the r th selected point in S be represented as s_r (appearing as the original r th point in S) and the $(r + 1)$ th selected point as $s_{(r+1)'}$ (appearing as the original $(r + 1)$ th point in S). The slope of Slp_r^S is calculated by the following formula:

$$Slp_r^S = (s_{(r+1)'} - s_r) / ((r + 1)' - r'). \quad (2)$$

If a selected point is not in the list of S (e.g. the 1.5th point), then a weighted average from its two adjacent points is used. The slopes are thus used to evaluate the distance of the two segments S and Q . The evaluation formula is listed as follows:

$$ED_{slopes}(Q, S) = \frac{\sqrt{\sum_{r=1}^{numSlopes} (Slp_r^Q - Slp_r^S)^2}}{numSlopes}, \quad (3)$$

where Q and S are two segments in a chromosome, Slp_r^Q and Slp_r^S are the r th slope values of Q and S . ED_{slopes} is then used as the clustering attribute to group the segments in a chromosome. An example is given below to illustrate the above idea.

Example 2. Continuing Example 1, assume the number of slopes is set at 2. Take segment S_1 , which is (8, 12, 12, 2), as an example. The first point in S_1 is 8 and the last point is 2. Since the number of slopes is 2, one more point is thus needed with the first and the last ones to form 2 slopes. The additional point is the 2.5th point, which has the value of $(12+12)/2 (=12)$. The three points to form the two slopes are thus 8, 12 and 2. The slope values of S_1 are thus 2.66 $(=(12-8)/(2.5-1))$ and $-6.66 (=(2-12)/(4-2.5))$. They are then represented as a pair (2.66, -6.66). In the same way, the slope values of S_2, S_3, S_4 and S_5 are calculated as (2, 2), $(-2, -2)$, (1.33, -1.33) and (1.33, -1.33), respectively. Also assume the parameter k in the k -means clustering approach is set at 2. Two clusters can thus be obtained. In this example, $cluster_1$ has the segment S_1 and $cluster_2$ has the segments S_2, S_3, S_4 and S_5 .

2.4. Incorporating discrete wavelet transformation in the proposed approach

The segments in a chromosome may have different length. Thus, an approach has to be designed to adjust them to the same length for calculating their similarities. In this paper, the *Discrete Wavelet Transformation* (DWT) is adopted for this purpose (Percival and Walden, 2000). Given an input time series which contains 2^n numbers, the Haar wavelet transform, which is the simplest wavelet transform, is used here to pair up the input values. Let T be a time series with n data points, represented as $T = \{d_1, d_2, \dots, d_k, \dots, d_n\}$. A set of coefficients and an approximation will then be generated in a single pass of the transform. The approximation is calculated as $(d_i + d_{i+1})/2$ and the coefficients as $(d_i - d_{i+1})/2$, for $i = 1, 3, 5, \dots, n - 1$. The approximation is then used as the input for the next pass of the wavelet transform. The process is executed recursively and totally $2^n - 1$ coefficients and an approximation will be finally derived.

Fig. 3 shows an example to illustrate the DWT process. Assume a time series T has eight data points, (2, 4, 6, 8, 8, 6, 4, 2), which is shown in Fig. 3a. After DWT, its results for different levels are shown in Fig. 3b–d.

In Fig. 3b, the approximation is $((2+4)/2, (6+8)/2, (8+6)/2, (4+2)/2)$, which is (3, 7, 7, 3), and the coefficients are $((2-4)/2, (6-8)/2, (8-6)/2, (4-2)/2)$, which are $(-1, -1, 1, 1)$ after the first wavelet transform. The results in the other two levels can be obtained in the same way. When the final results shown in Fig. 3d have $2^3 - 1 (=7)$ coefficients, the wavelet transform stops. Each wavelet transform will generate an approximation with a half length of the original one. With this property, DWT is used here to adjust segments such that they can have the same length for calculating their similarities. However, in some cases, the adjusted length of the longer segment may still not match the length of the shorter

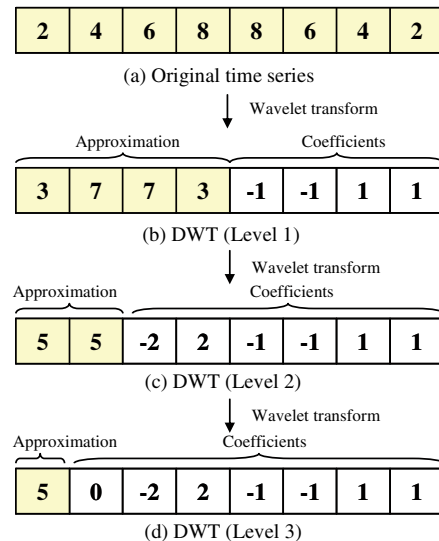


Fig. 3. The DWT process in the example.

segment. For example, assume there are two segments and their lengths are 31 and 64, respectively. The length of the longer segment will first be reduced to 32. Since the length of the shorter segment is 31, only the first 31 data points from the longer segment will be used to calculate the similarity between the two segments.

2.5. Fitness and selection

In order to get a good segmentation result from an initial population, the genetic algorithm selects good *parent* segmentation results for mating and mutation to generate the next generation. Here, the Roulette-wheel selection and the elitist selection strategies can be used for generating a better next generation. An evaluation function is also needed to qualify the derived segmentation results. Before the fitness function is described, several related terms are first defined below.

The distance value $D(seg_{qj}^h)$ of the j th segment in the h th cluster of a chromosome C_q is defined as follows:

$$D(seg_{qj}^h) = \min_{g \neq j} \{ \text{Dist}(seg_{qj}^h, seg_{qg}^h) \}, \quad (4)$$

where $\text{Dist}(seg_{qj}^h, seg_{qg}^h)$ is the Euclidean distance of the two segments seg_{qj}^h and seg_{qg}^h by using formula (1). When seg_{qj}^h and seg_{qg}^h have different lengths, the discrete wavelet transformation (DWT) is applied to the longer segment to adjust its length to that of the shorter one. The two transformed segments, instead of the original ones, are thus compared to calculate the distance between them. Thus instead of the parameter n in formula (1), the minimum length of segments seg_{qj}^h and seg_{qg}^h is used.

Let $D(seg_{qj}^h)$ denote the minimum Euclidean distance between seg_{qj}^h and any other segments in the same h th cluster. The segment with the minimum distance to seg_{qj}^h can also be thought of as most similar to it. Next, the distance $\text{ClusterD}(cluster_q^h)$ of the h th cluster in a chromosome C_q is defined as follows:

$$\text{ClusterD}(cluster_q^h) = \sum_{j=1}^{sn_q^h} D(seg_{qj}^h), \quad (5)$$

where sn_q^h is the number of segments in the cluster $cluster_q^h$ in C_q . The value ClusterD can thus represent the similarity degree of the segments in a cluster. If there is only one segment existing in a cluster, then a default distance is assigned to the cluster. Below, an example is given to illustrate the above concepts.

Example 3. Continue Example 2, where the chromosome $C_1 = \{4, 8, 12, 16\}$. The five segments are divided into two clusters, with S_1 alone belonging to the first cluster and the other four segments belonging to the second cluster. Assume the default distance for one-segment cluster is 4. The fitness value of C_1 is calculated as follows.

$$\begin{aligned} f(C_1) &= \text{ClusterD}(cluster_1^1) + \text{ClusterD}(cluster_1^2) \\ &= D(seg_{11}^1) + D(seg_{11}^1) + D(seg_{12}^2) + D(seg_{13}^2) + D(seg_{14}^2) \\ &= 4 + 1.58 + 1.58 + 0 + 0 = 7.16. \end{aligned}$$

2.6. Genetic operators

Genetic operators are important to the success of specific GA applications. Two genetic operators, the *one-point crossover* and the *one-point mutation*, are used in the proposed algorithm as in Chung et al. (2004). The one-point crossover operator will create two offspring chromosomes from two parent ones by randomly choosing a cut point. For example, assume there are two parent

chromosomes $C_1 = \{4, 8, 12, 16\}$ and $C_2 = \{4, 12\}$ to be mated as shown in Fig. 4a. If the cut point is set at 12, then the results after crossover are $C'_1 = \{4, 8, 12\}$ and $C'_2 = \{4, 12, 16\}$ as shown in Fig. 4b.

The one-point mutation operator will create a new segmentation result by adding a cut point into or removing a cut point from a chromosome. Adding or removing a cut point is decided with probabilities. For example, assume the cut point 12 is removed from the chromosome C_1 , which is $\{4, 8, 12, 16\}$. The result C'_1 will become $\{4, 8, 16\}$ as shown in Fig. 5a. If the cut point 2 is added into the chromosome C'_1 , the result C''_1 will become $\{2, 4, 8, 16\}$ as shown in Fig. 5b.

3. The proposed time-series segmentation algorithm

According to the above description, the proposed time-series segmentation algorithm is stated as follows.

3.1. The time-series segmentation algorithm

INPUT: A time series T with n data point, a parameter k (*PatternNum*) for k -means clustering, a population size $PSize$, a crossover rate P_c , a mutation rate P_m , a parameter $numSlopes$ for the slope number of a segment, a minimum length $MinLength$, and a DWT compression level $CWTLevel$.

OUTPUT: A segmentation result with k patterns.

STEP 1: Generate an initial population of $PSize$ individuals according to the method mentioned in Section 3. Each individual is a possible time-series segmentation result.

STEP 2: Divide the segments of a chromosome into k clusters by the following substeps.

SUBSTEP 2.1: Calculate the $numSlopes$ slopes of each segment in a chromosome as mentioned in Section 2.3. Each segment S is thus transformed as a slope tuple of $(Slp_1^S, Slp_{2,...,numSlopes}^S)$.

SUBSTEP 2.2: Randomly select k segments as the initial centers.

SUBSTEP 2.3: Assign each segment S to one of the k groups according to the Euclidean distance of the slope values of the segment S and the centers by using formula (2).

SUBSTEP 2.4: Calculate the new center of each group by averaging the tuple $(Slp_1^S, Slp_{2,...,numSlopes}^S)$ of segments in the same group.

SUBSTEP 2.5: If the new centers do not change, go to the next step; otherwise, go to SUBSTEP 2.3.

STEP 3: Calculate the fitness value of each chromosome C_q by the following formula:

$$f(C_q) = \sum_{h=1}^k \text{ClusterD}(cluster_q^h).$$

STEP 4: Execute the crossover operation on the population.

STEP 5: Execute the mutation operation on the population.

STEP 6: Use the selection operation to choose appropriate individuals for the next generation. Here, the selection operator can be the Roulette-wheel or the elitist strategy.

STEP 7: If the termination criterion is not satisfied, go to Step 2; otherwise, output the best chromosome.

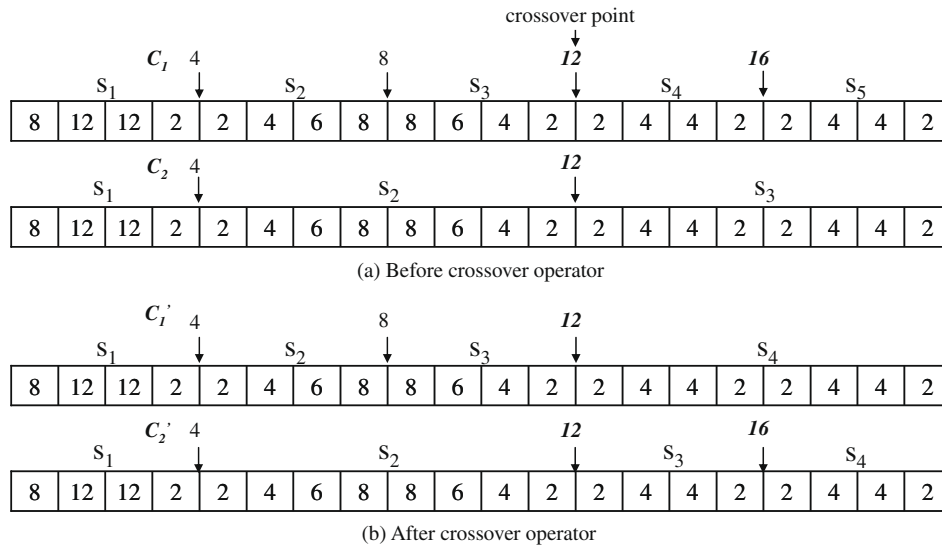


Fig. 4. The crossover operation.

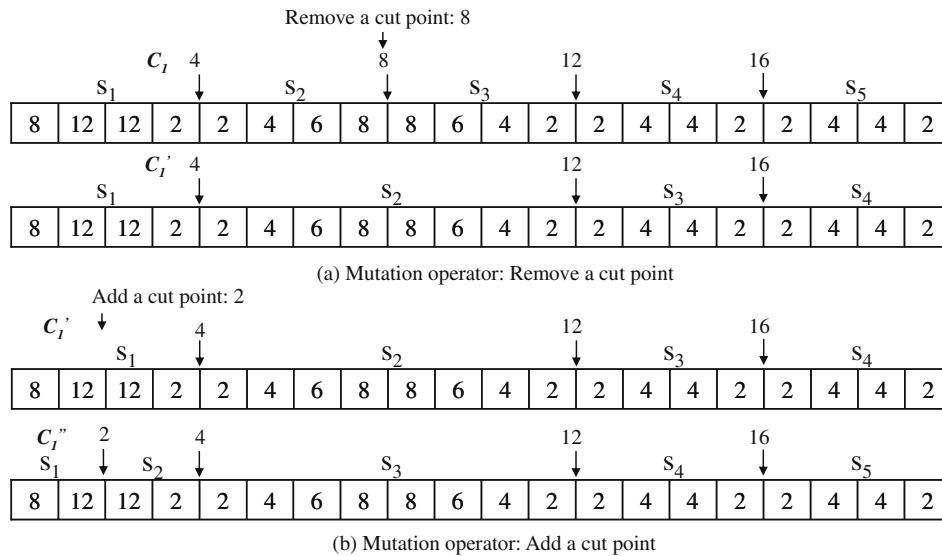


Fig. 5. The mutation operation.

Table 2

The parameters used in the experiments.

Parameter	Description	Value
Generation	Number of iterations	500
PSize	Number of chromosomes	80
PatternNum	Number of clusters	6
numSlopes	A number of slopes of a segment	2
MinLength	Minimum length	4 (Length = $2^4 = 16$)
DwtLevel	The most level that DWT can be compressed	2
P_c	Crossover rate	0.8
P_m	Mutation rate	0.3

4. Experimental results

In this section, experiments made to show the performance of the proposed approach are described. They were implemented in Java on a personal computer with Intel Pentium IV 3.20 GHz

and 512MB RAM. A real financial dataset from of steelmakers in Taiwan was used to demonstrate the proposed approach. It contained the stock prices from January 1, 2005 to December 31, 2006. The parameters used in the experiments are shown in Table 2.

In order to verify the segmentation results, the known financial patterns defined in Chung et al. (2004) are first shown in Fig. 6.

There are 24 financial patterns in Fig. 6. These financial patterns are then used to verify the derived segmentation results. After 500 generations, the final best chromosome from the dataset is shown in Fig. 7.

In Fig. 7, the number marked on top of a segment represents the cluster that the segment belongs to. When compared with the financial patterns defined in Fig. 6 (Chung et al., 2004), the segmentation results contain four financial patterns, including *Down-trend*, *Spike Bottom*, *Rounded top* and *Uptrend*, respectively for the second, the third, the fourth and the fifth clusters. Although the

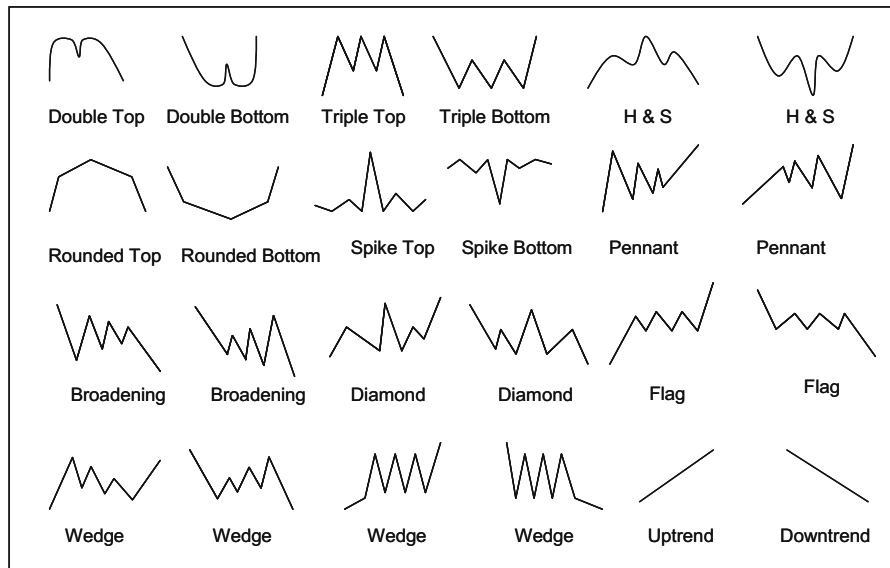


Fig. 6. The known financial patterns.

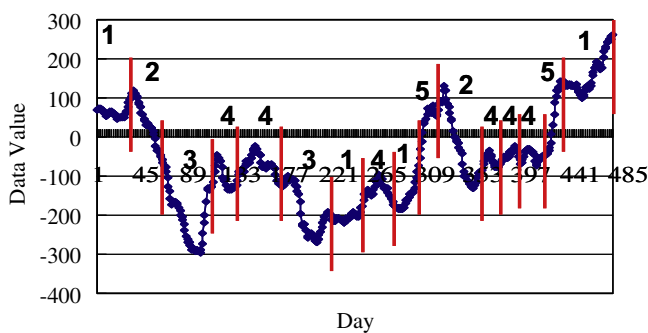


Fig. 7. The segmentation results by the proposed approach.

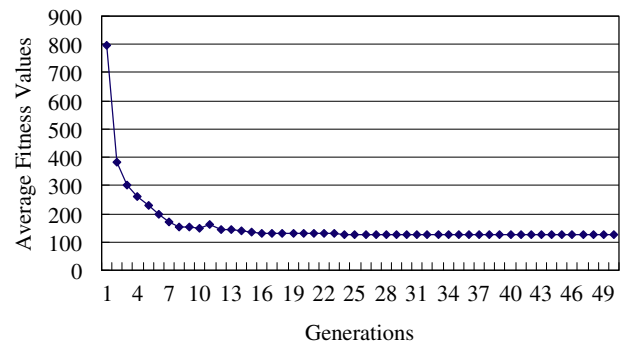


Fig. 8. The convergence results of the proposed approach.

remaining segments in the first cluster are an un-known pattern, it can still be provided as a new pattern for further analysis.

Note that in the experiment, the parameter *PatternNum* was set at six. However, from Fig. 7, it can be observed that the proposed approach derived only five patterns. This phenomenon was caused due to a limitation of the *k*-means clustering technique, which is called degeneracy (Jain et al., 1999). Other limitations of the *k*-means clustering technique, like how to process outlier, generate the initial seeds and decide the clustering number, have also been described in Jain et al. (1999). Some deeper analysis and discussion about them can be found in Kuncheva and Vetrov (2006) and Li et al. (2008). Because the *k*-means clustering algorithm is very dependent on initialization, the dataset were thus run for ten times with the best result kept to alleviate the local minima problem. The convergence results for one run are shown in Fig. 8.

It can be observed from Fig. 8 that the average fitness values gradually converged along with the increase of generations as expected. The above experimental results show that the proposed approach is effective no matter in finding patterns or convergence. Besides, the derived segmentation results by the proposed approach are easy to be applied to applications. For example, the proposed approach can be used to segment stock prices of other steelmakers. Segmentation results and their related patterns are then used for further analysis, such as discovering common patterns or associations between patterns.

The proposed approach was then compared to the PIP (Perceptually Important Points) module used for segmentation in (Chung et al., 2002, 2004; Feng et al., 2005). The results by the PIP module are shown in Fig. 9.

It could be seen from the figure that nearly all the segment points by PIP fell at local maxima or minima in the time series. It was thus hard to judge a variety of patterns out because only two patterns (from up to down and from down to up) could be found in the figure. Thus, it had to be combined with other analysis

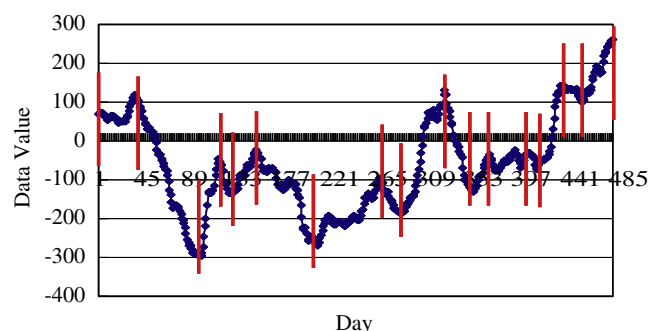


Fig. 9. The segmentation results by using the PIP module.

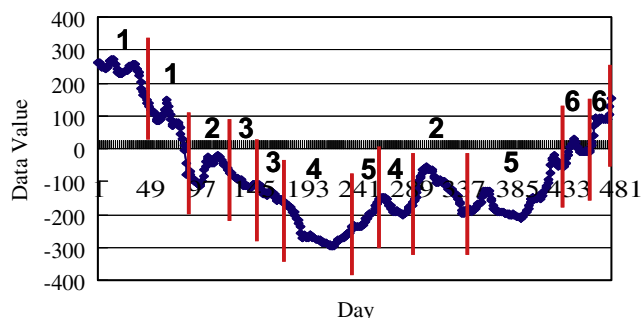


Fig. 10. The segmentation results for the second data set.

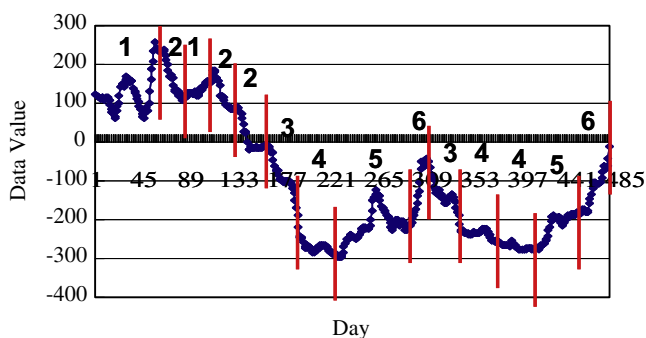


Fig. 11. The segmentation results for the third data set.

techniques to further find appropriate patterns. Instead, our approach could segment a time series and found appropriate patterns at the same time.

The proposed approach was then compared to Chung et al. (2004), which also used GA to find patterns. Their approach needed to define the desired patterns in advance, but ours did not. Chung et al.'s approach is thus very suitable for the applications in which users assure what patterns they want. Instead, our approach is suitable for the applications in which users do not know the patterns.

More experiments on other two financial datasets of steelmakers in Taiwan were made to further verify the effectiveness of the proposed approach. Experimental results are shown in Figs. 10 and 11. It is seen from the figures that the given time series could be appropriately segmented by the proposed approach.

5. Conclusion and future works

In this paper, we have proposed a time-series segmentation algorithm by using the clustering technique, the discrete wavelet transformation and the genetic algorithm. The genetic algorithm is used to find the segmentation points for deriving appropriate patterns. In the fitness evaluation, the proposed algorithm first divides subsequences in a chromosome into k clusters by using the k -means clustering approach. The Euclidean distance is used to calculate the distance of each segment and to evaluate the fitness value of a chromosome. The discrete wavelet transformation is also used to adjust the length of the subsequences since the length of two subsequences may be different. Real datasets are also used to show the performance of the proposed approach.

The proposed approach can not only find the segmentation result but can also derive appropriate patterns from the time series by combining the genetic algorithms and the clustering tech-

niques. The derived patterns can also be scaled up via the discrete wavelet transformation. These properties thus make the derived segments and patterns more flexible and more diverse. It may thus help experts easily find their desired interesting results for further analysis. In the future, we will continuously attempt to enhance the time-series segmentation approach for more complex problems.

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